PEMDAS REFERENCE SHEET

Name: ..... **Date:** .....

The order of operations: Parentheses, Exponents, Multiplication and Division (left to right), Addition and Subtraction (left to right). Keep this page in your folder.

Step	Do this	Example
P	Work inside **parentheses** first	$(3 + 5) \times 2 \rightarrow 8 \times 2$
E	**Exponents** (squares, cubes)	$4^2 + 1 \rightarrow 16 + 1$
M D	**Multiply and divide**, left to right (neither comes first)	$12 \div 3 \times 2 \rightarrow 4 \times 2 \rightarrow 8$
A S	**Add and subtract**, left to right	$10 - 4 + 3 \rightarrow 6 + 3 \rightarrow 9$

THREE WORKED LADDERS

$7 + 3 \times 4$
$= 7 + 12$
$= 19$

$(9 - 4) \times 3 + 2$
$= 5 \times 3 + 2$
$= 15 + 2$
$= 17$

$2^3 + 12 \div 4$
$= 8 + 12 \div 4$
$= 8 + 3$
$= 11$

COMMON MISTAKES

- Working left to right no matter what: $2 + 3 \times 4$ is 14, not 20.
- Always multiplying before dividing: M and D are one step, left to right. $18 \div 3 \times 2 = 12$.
- Always adding before subtracting: A and S are one step too. $10 - 4 + 3 = 9$.
- Squaring the wrong thing: 2×3^2 means $2 \times 9 = 18$, not 6^2.

QUICK CHECKS: TRUE OR FALSE?

1. $8 - 2 + 3 = 9$ T / F
2. $2 + 3 \times 4 = 20$ T / F
3. $(2 + 3) \times 4 = 20$ T / F
4. $18 \div 3 \times 2 = 3$ T / F
5. $5^2 - 10 = 15$ T / F
6. $10 - 4 \div 2 = 3$ T / F

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PEMDAS REFERENCE SHEET ANSWER KEY

Name: ..... Date: .....

Answer key for the quick checks.

Step	Do this	Example
P	Work inside **parentheses** first	(3 + 5) × 2 → 8 × 2
E	**Exponents** (squares, cubes)	4^2 + 1 → 16 + 1
M D	**Multiply and divide**, left to right (neither comes first)	12 ÷ 3 × 2 → 4 × 2 → 8
A S	**Add and subtract**, left to right	10 − 4 + 3 → 6 + 3 → 9

THREE WORKED LADDERS

7 + 3 × 4
= 7 + 12
= 19

(9 − 4) × 3 + 2
= 5 × 3 + 2
= 15 + 2
= 17

2^3 + 12 ÷ 4
= 8 + 12 ÷ 4
= 8 + 3
= 11

COMMON MISTAKES

- Working left to right no matter what: 2 + 3 × 4 is 14, not 20.
- Always multiplying before dividing: M and D are one step, left to right. 18 ÷ 3 × 2 = 12.
- Always adding before subtracting: A and S are one step too. 10 − 4 + 3 = 9.
- Squaring the wrong thing: 2 × 3^2 means 2 × 9 = 18, not 6^2.

QUICK CHECKS: TRUE OR FALSE?

1. 8 − 2 + 3 = 9 **true**
2. 2 + 3 × 4 = 20 **false**
3. (2 + 3) × 4 = 20 **true**
4. 18 ÷ 3 × 2 = 3 **false**
5. 5^2 − 10 = 15 **true**
6. 10 − 4 ÷ 2 = 3 **false**

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